The Severity of each Condition

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I believe that it's important to understand how serious medical conditions are

Statistics

Brief Summary: This dataset shows how different hospitals perform better than others based on both mortality rate and quality throughout the years.

This analysis can be seen through the following categories:

- County
- Hospital
- Cases of each condition
- Deaths of patients with each condition
- Type of medical condition

Anomalies: Found and threw <null>, ‘,’, and ‘.’ in numeric categories.

Cases vs Deaths

Comparing the Data for the following the conditions:

- Pneumonia: 4117
- Heart Failure: 4068
- GI Hemorrhage: 3851

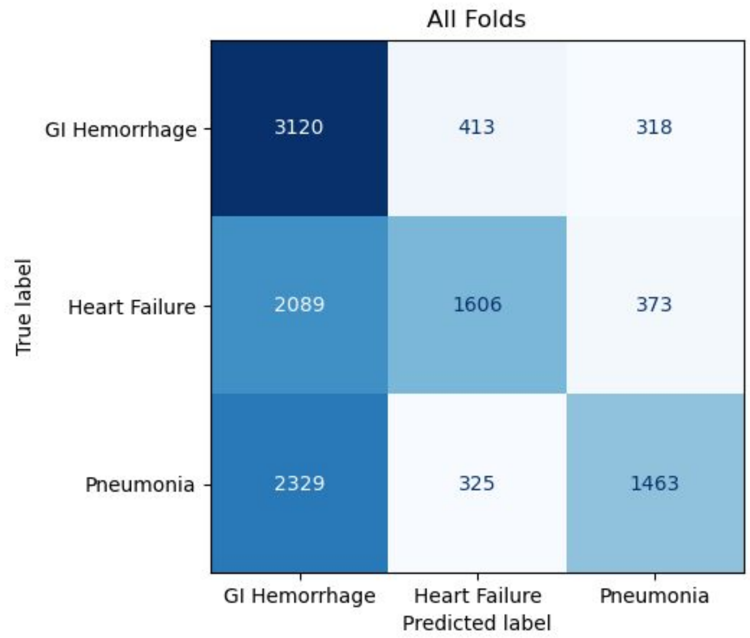
Each dot is unique with the year, hospital, and number of cases

K-Fold Cross-Examination

	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Overall
Accuracy	0.519	0.514	0.506	0.514	0.518	0.514
Balanced	0.524	0.520	0.512	0.520	0.524	0.520

Used K-fold to split the data into 5 different folds

Used a standard scaler to scale the features in X



Confusion Matrix

Very inaccurate due to the imbalance of data per condition:

Pneumonia: 4117

Heart Failure: 4068

GI Hemorrhage: 3851

Conclusion

Overall accuracy: 51%

Future experiments: Will try to balance the data to a 1:1 ratio to maximize accuracy

May want to dig deeper into this type of data and use a more broader dataset with more features for comparison and prediction